Answer **all** the questions in this section in the spaces provided

1 A rigid uniform beam of weight W is connected to a fixed support by a hinge as shown in Fig. 1.1.

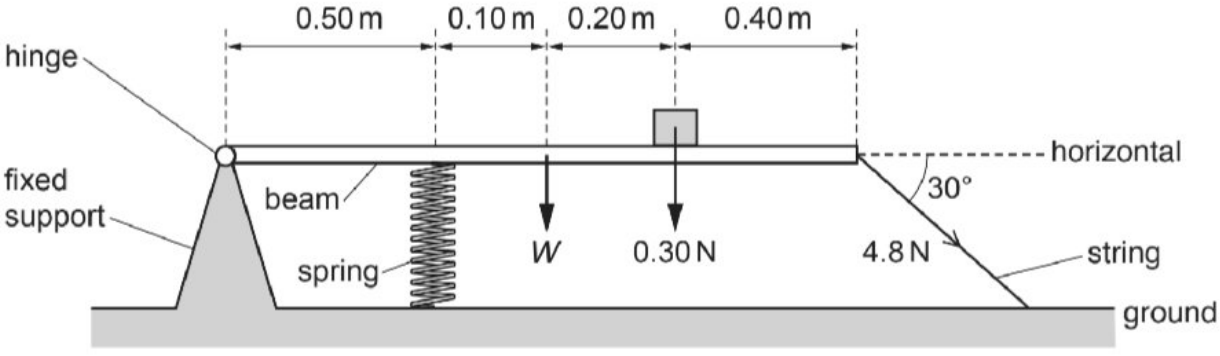


Fig. 1.1 (not to scale)

A compressed spring exerts a vertical force of 8.2 N on the horizontal beam. A block of weight 0.30 N rests on the beam. The right end of the beam is connected to the ground by a string at an angle of 30° to the horizontal. The tension in the string is 4.8 N. The distances along the beam are shown in Fig. 1.1.

The beam is in equilibrium. Assume that the hinge is frictionless.

(a) (i) Determine the weight W of the beam.

W = ... N [3]

(ii) Calculate the horizontal force exerted on the beam by the hinge.

force = ... N [1]

(b) The compressed spring obeys Hooke's law and stores an elastic potential energy of 0.32 J.

Calculate the compression of the spring.

compression = ...m [2]

[Total: 6]